DEYANG HSU

deyangh.github.io

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EDUCATION

University of Southern California — Los Angeles, California

Master's in Computer Science, Viterbi School of Engineering — August 2024-May 2026

The Ohio State University — Columbus, Ohio

Bachelor of Science in Electrical Computer Engineering — August 2018-May 2022

Honors: Magna Cum Laude | Dean's List Scholar | Alumni Society Scholarship recipient — GPA: 3.769/4.0

Relevant Coursework:

Machine Learning & AI | OOP | Discrete Structures | Data Structures and Algorithms |Computer Architecture | Parallel Computing | Database systems | Operating Systems | Web Crawling and Search Engines | LLM | NLP | Cybersecurity | Computer Networking | Multimodal Probabilistic Learning | Natural Language Dialogue Systems

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, HTML

Technologies & Tools: TensorFlow, scikit-learn, XGBoost, LightGBM, Pandas, NumPy, HuggingFace, NLTK, OpenMP, CUDA, SLURM, Multithreading, MPI, Relational Databases, Web Crawling, RAG

Systems: RISC-V, CISC, GitHub, Subversion, Networking, Operating Systems, Computer Architecture

RESEARCH EXPERIENCE

Research Assistant, FORTIS Lab | University of Southern California — January 2025 – Present

- Conducted research within the FORTIS (Foundations Of Robust Trustworthy Intelligent Systems) Lab under **Professor Zhao Yue**, focusing on developing robust, trustworthy, scalable AI systems emphasizing anomaly detection, out-of-distribution detection, structured data modeling, graph learning, and generative AI.
- Executed research on Comparative Narrative Analysis (CNA), focusing on TELeR taxonomy levels 1 to 4, leveraging large language models (GPT-4, DeepSeek, Claude) to perform evaluation and analytical tasks across multiple narratives.
- Improved model evaluation methodologies and automated assessment processes for CNA tasks, specifically narrative conflict detection, overlaps, uniqueness, and holistic summarization.

PROJECTS

Customer Churn Prediction Model

- Built a comprehensive machine learning pipeline to predict customer churn using scikit-learn, XGBoost, and LightGBM.
- Executed rigorous feature engineering and data preprocessing techniques, including handling class imbalance, feature scaling, and encoding categorical variables.
- Utilized cross-validation and hyperparameter tuning (GridSearchCV and RandomizedSearchCV) to achieve an optimized predictive accuracy, improving the baseline performance significantly.
- Presented insights and actionable strategies derived from model interpretations to reduce customer churn effectively.

WORK EXPERIENCE

Intel — Chandler, Arizona

Software Development Engineer — June 2022-August 2023

- Spearheaded the resolution of issues in-house software services, resulting in a 27% increase in software uptime
- Streamlined internal production database systems using PL/SQL, significantly boosting reliability and performance
- Orchestrated the validation of software, middleware services, and server clusters across semiconductor fabs
- Developed and deployed custom production software solutions, leading to an 18% rise in wafer production efficiency

General Motors — Bay City, Michigan

Manufacturing Controls Engineer Intern — June 2021-August 2021

- Troubleshot and optimized robotics manufacturing systems and PLCs, reducing production downtime by 8%
- Designed and implemented an IO-Link pressure sensor system with Siemens SIMATIC STEP 7, decreasing inventory scrapping by 13%